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Faculty of Computing

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SECP2613 System Analysis and Design

Phase 3 – Analysis and Design

Section 01

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Logical DFD (to be)

Context diagram

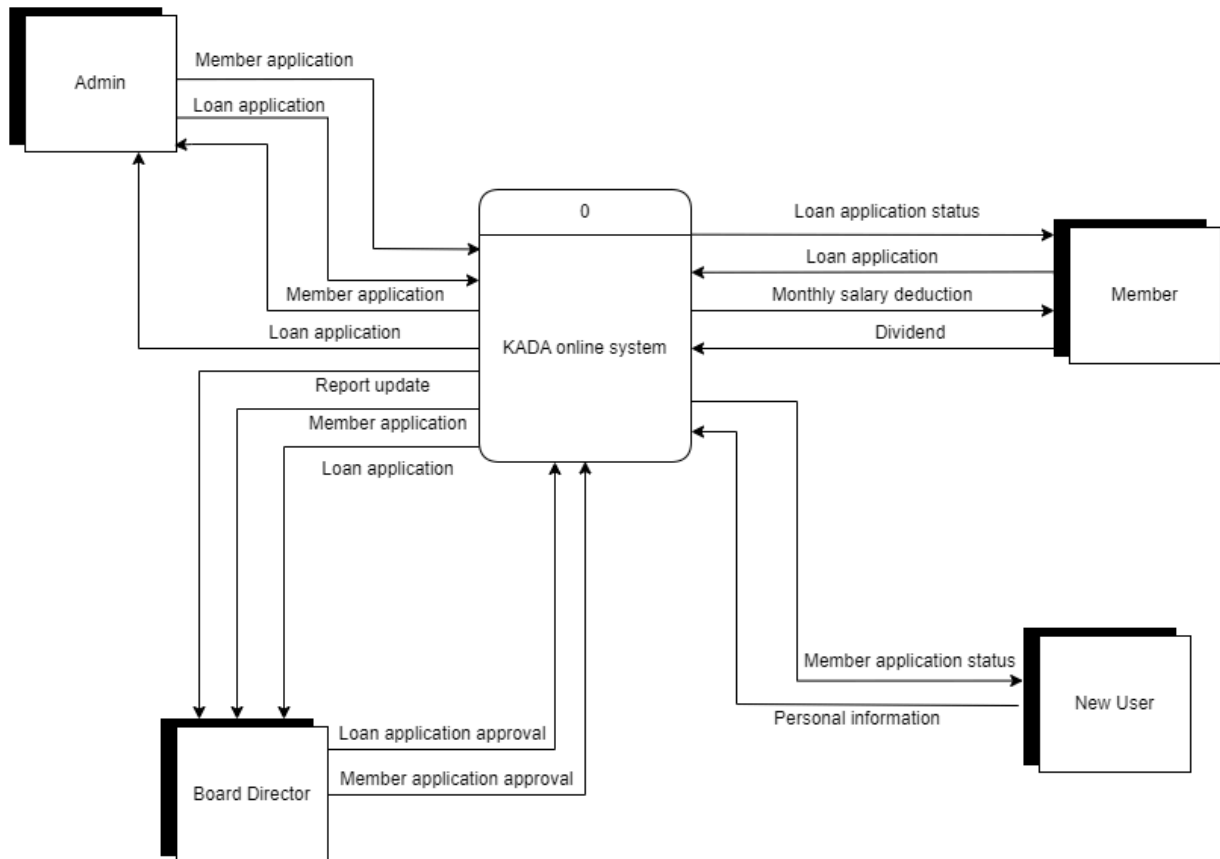
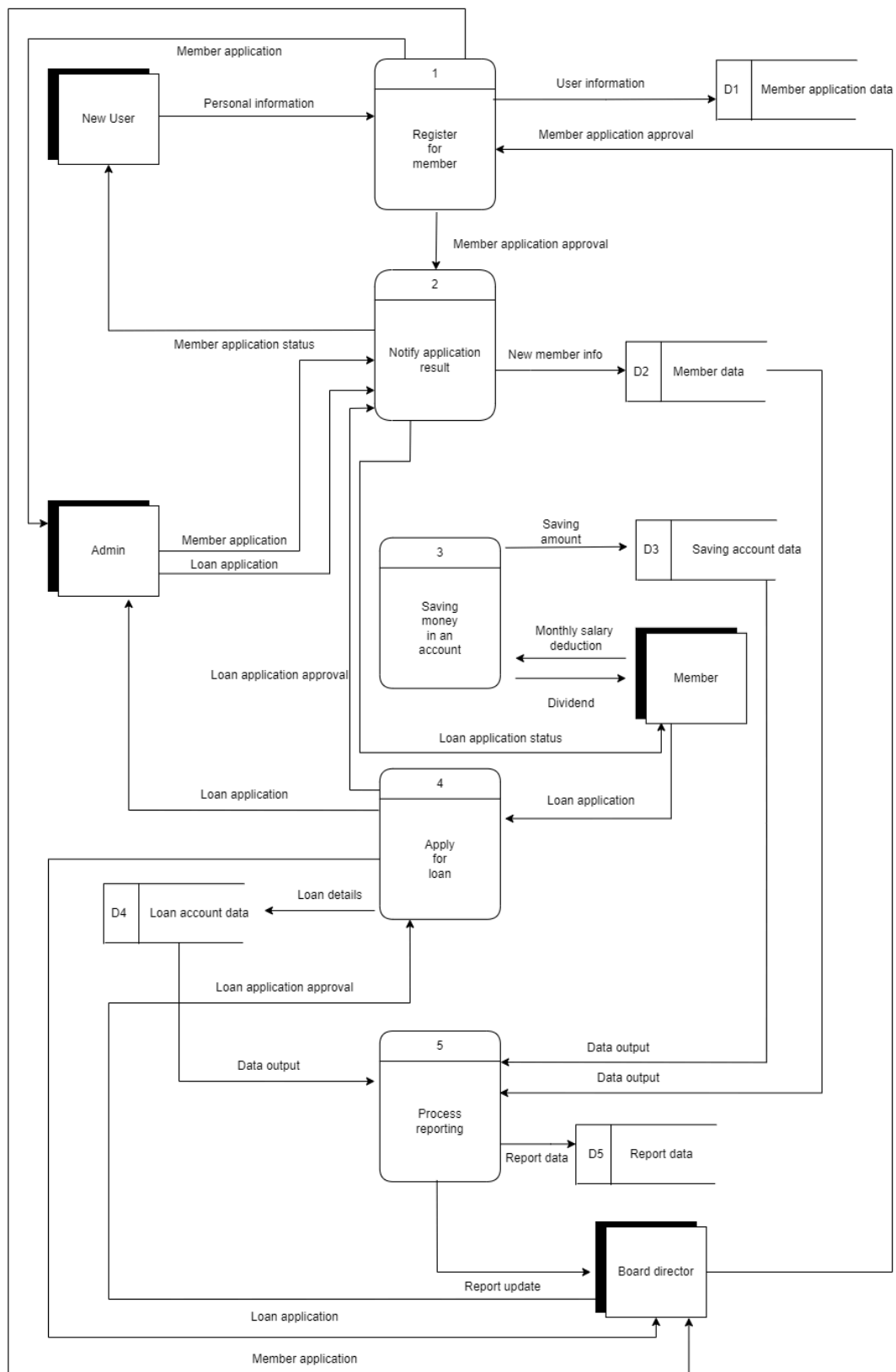
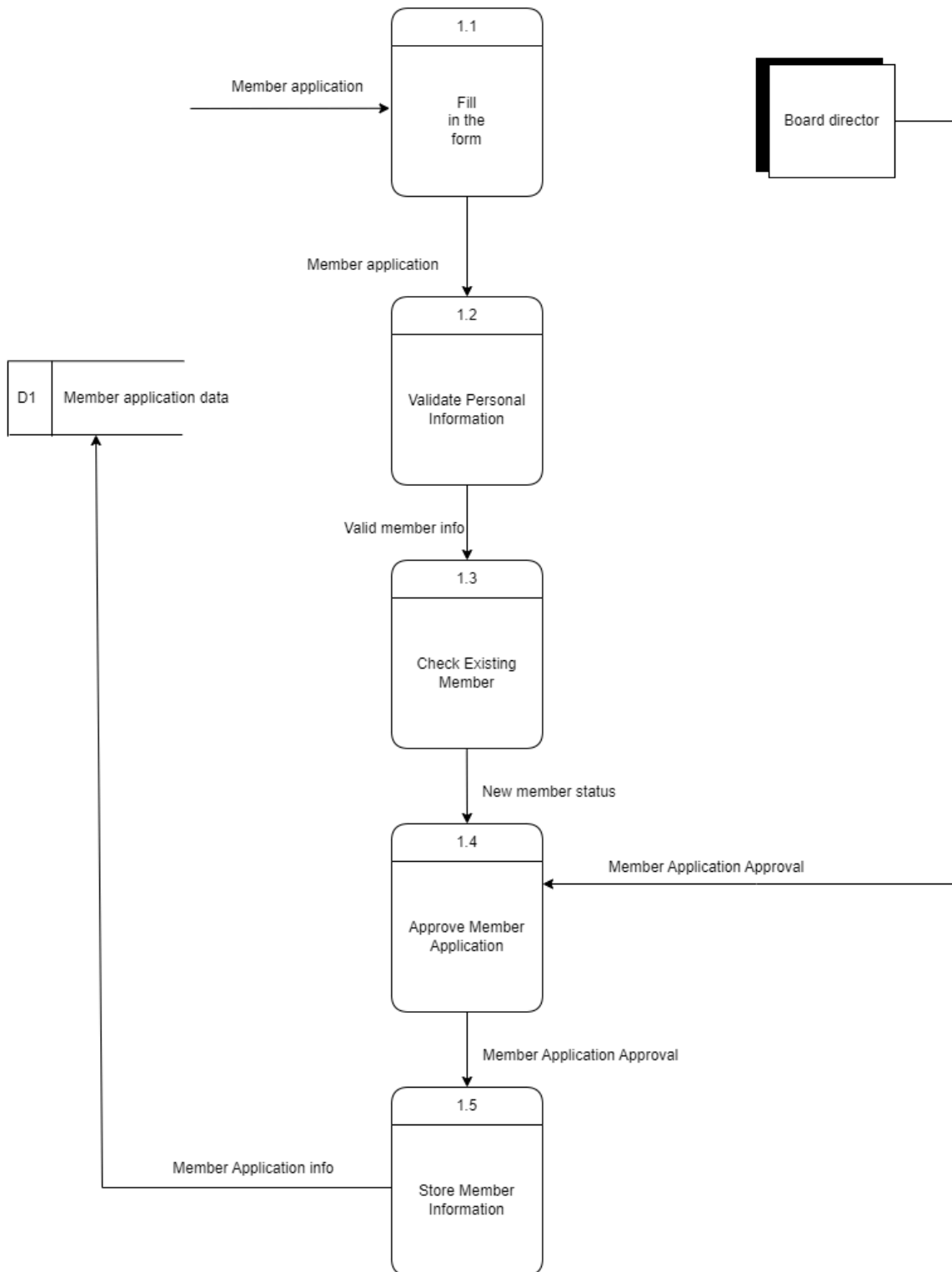


Diagram 0

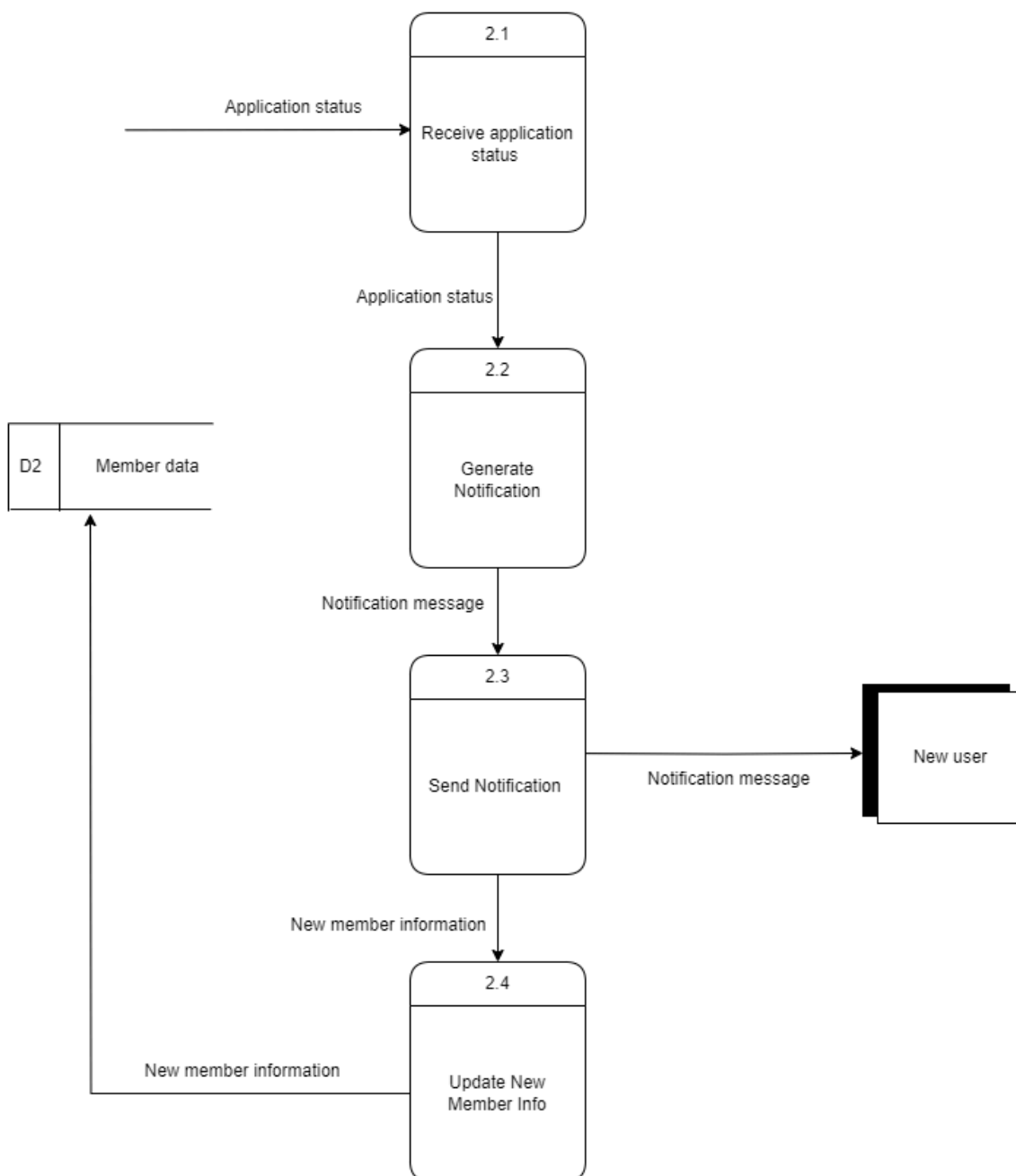


Child diagram

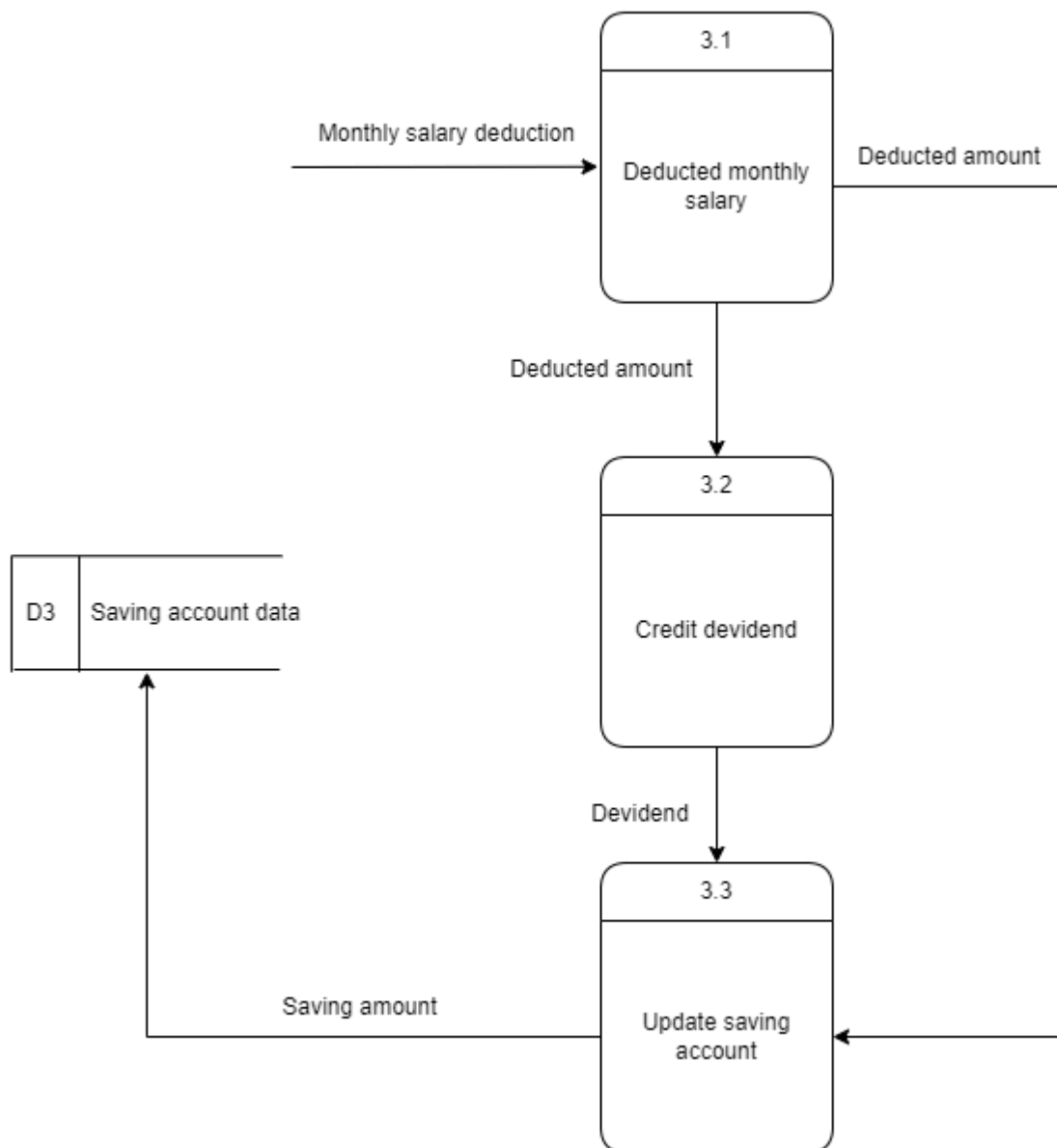
1.0 Register for member



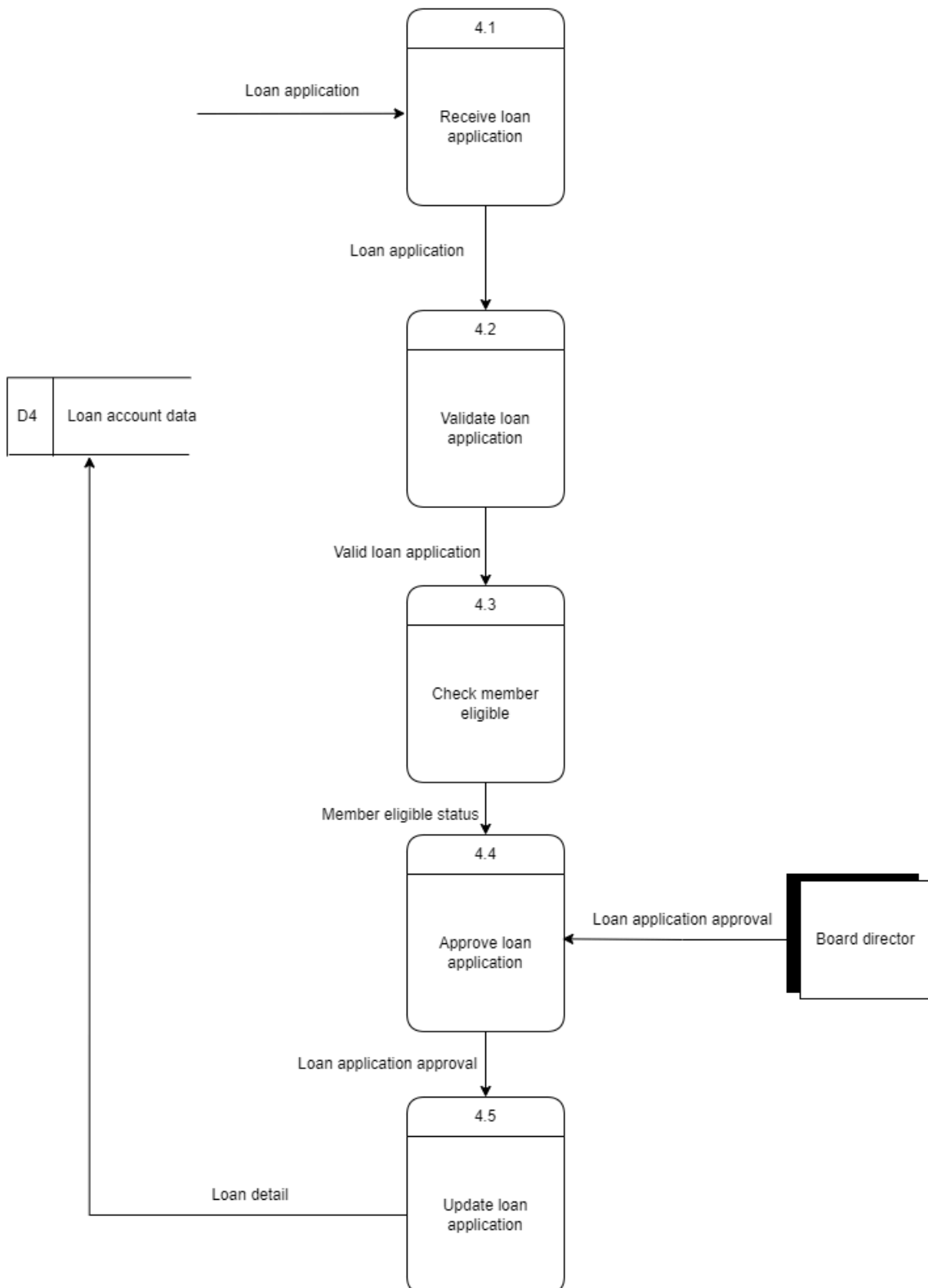
2.0 Notify application result



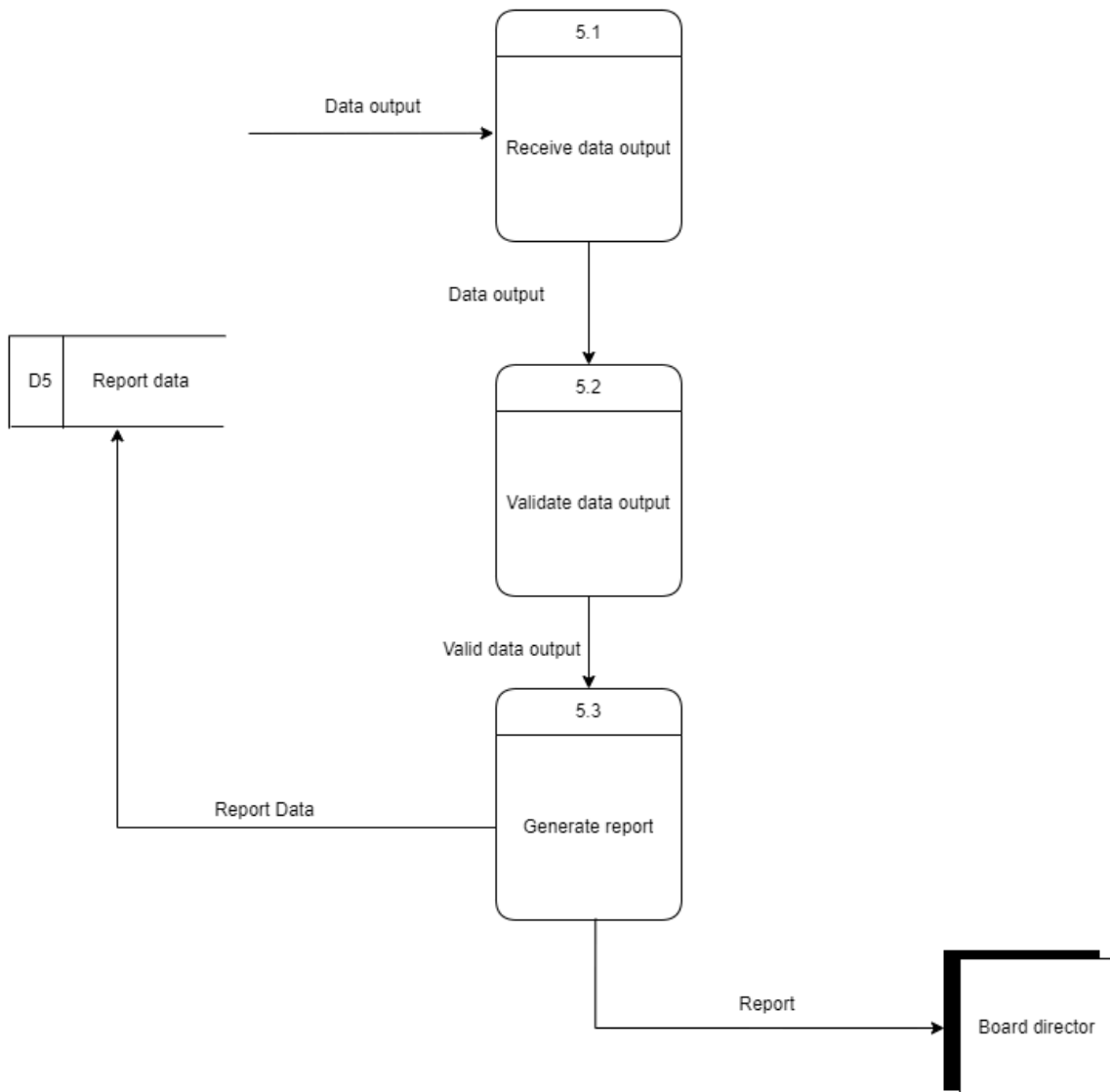
3.0 Saving money in an account



4.0 Apply for loan

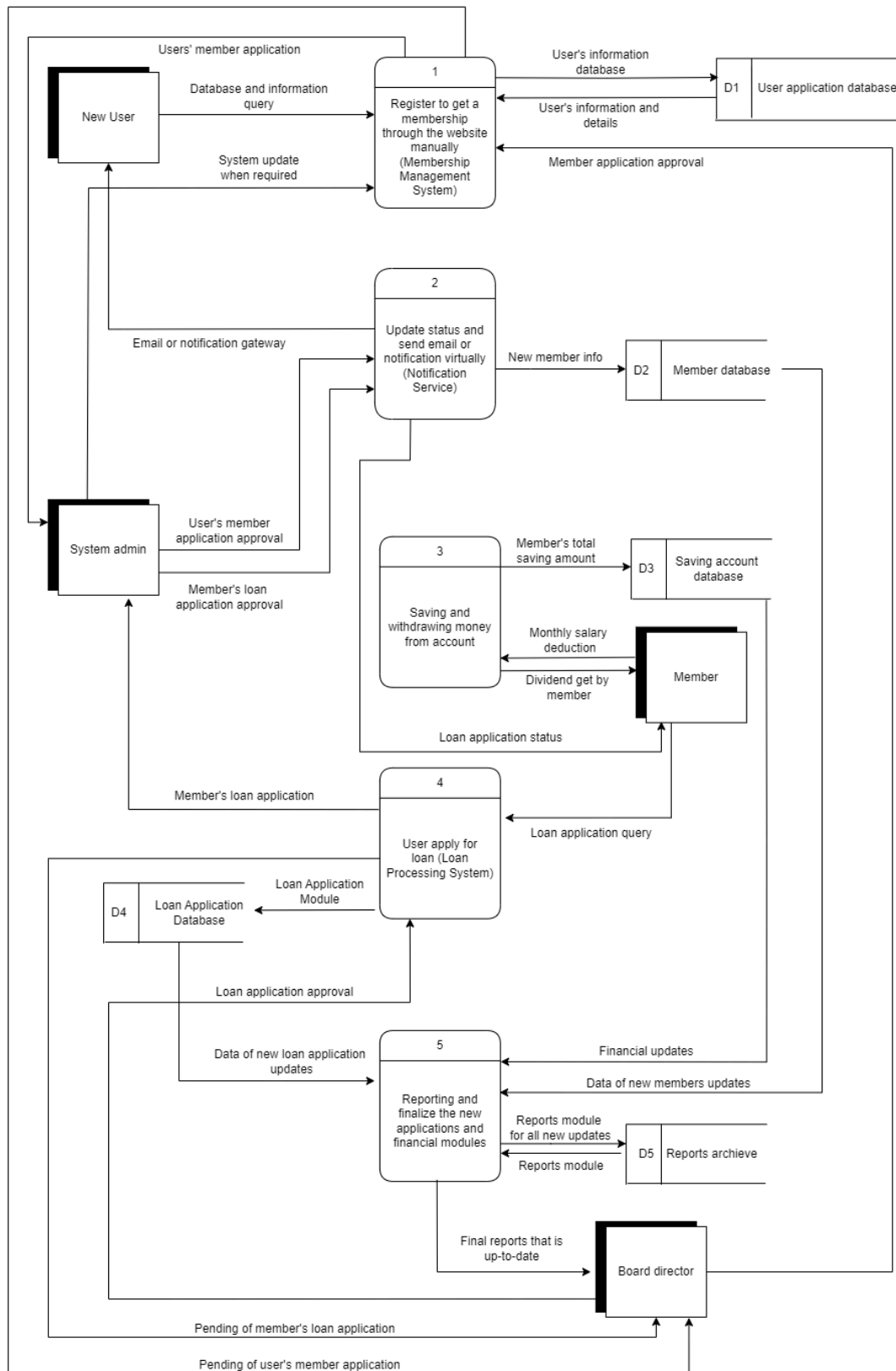


5.0 Process reporting



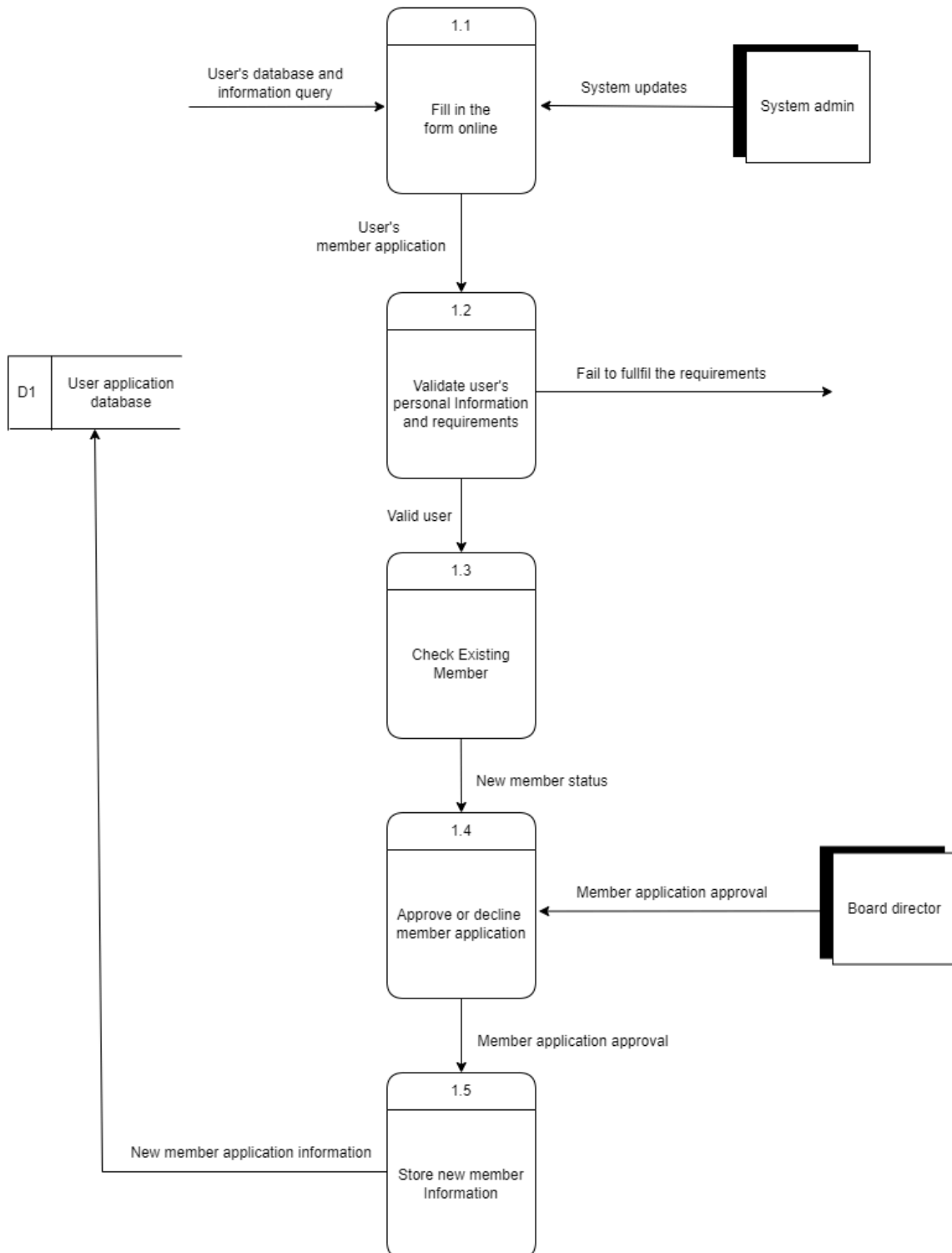
Physical DFD

Diagram 0

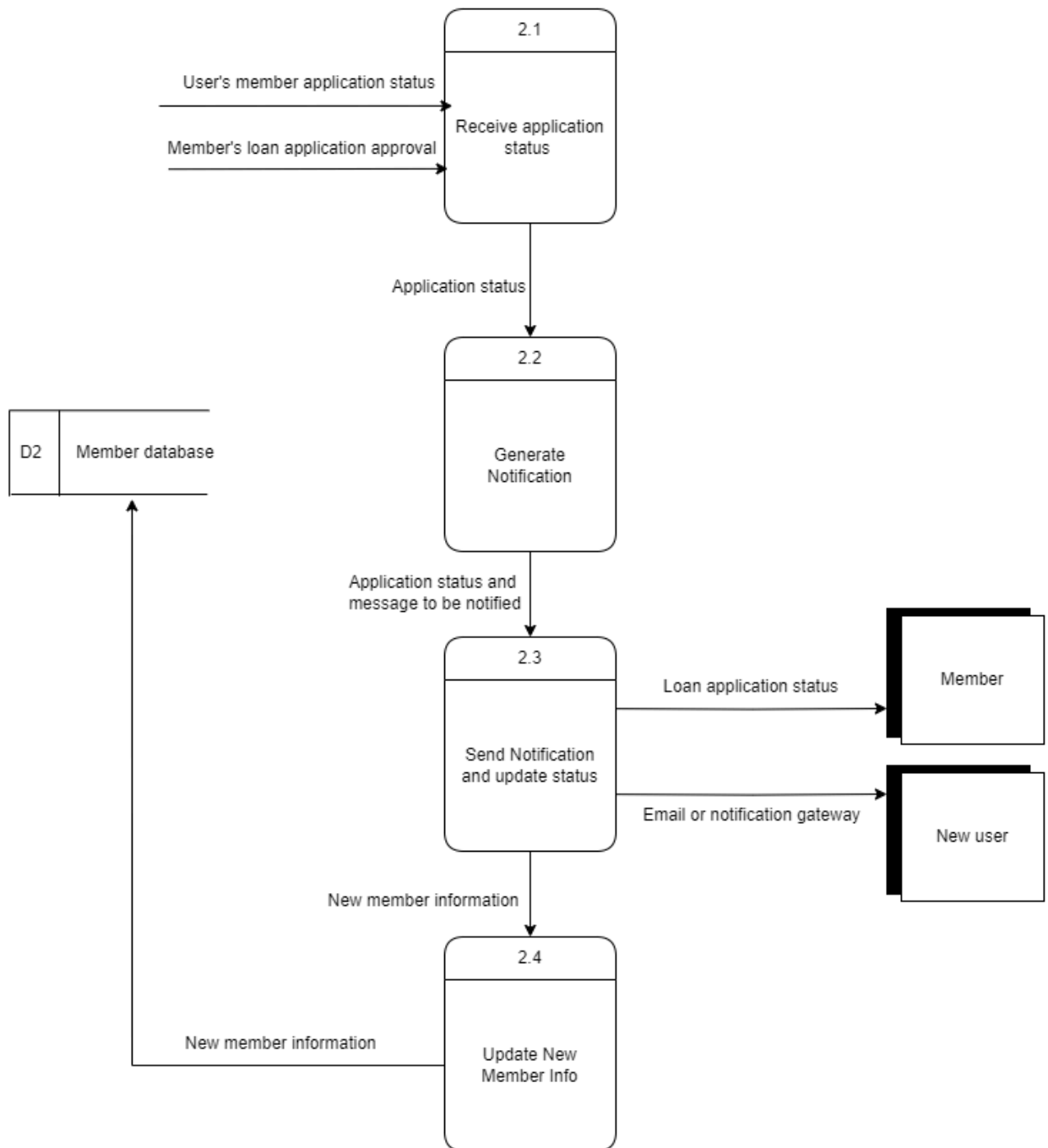


Child diagram

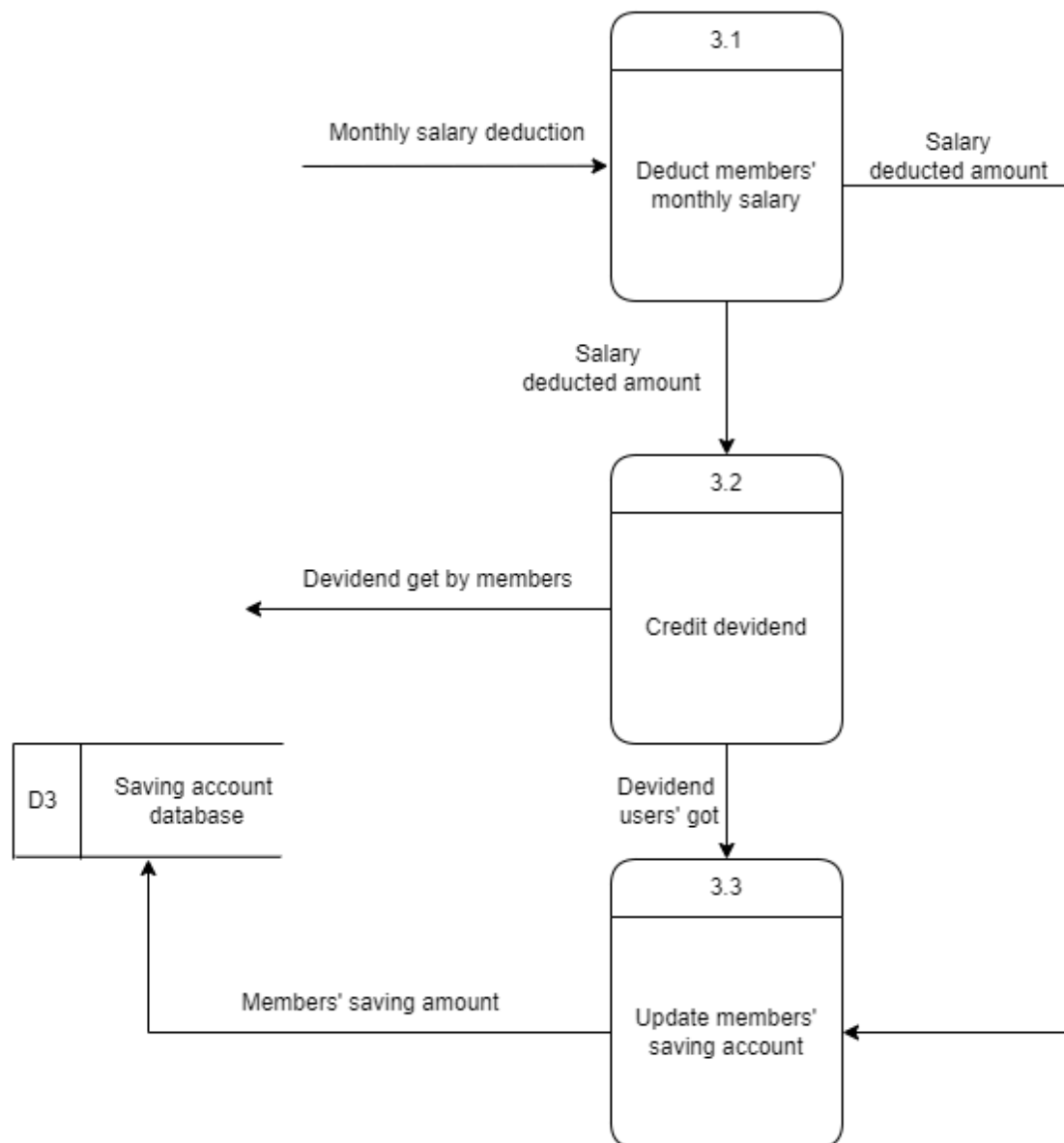
1.0 Register to get a membership through the website manually (Membership Management System)



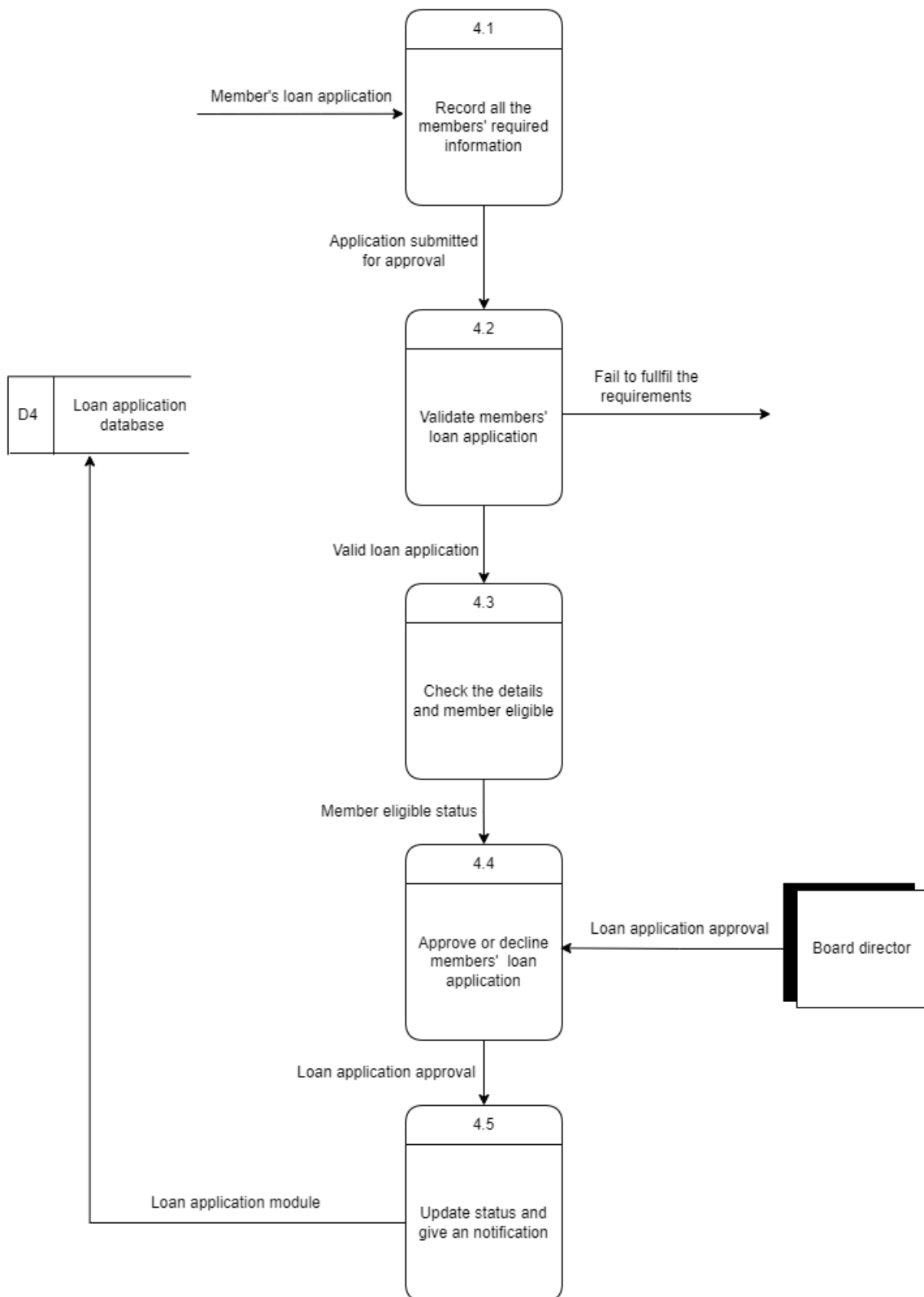
2.0 Update status and send email or notification virtually (Notification Service)



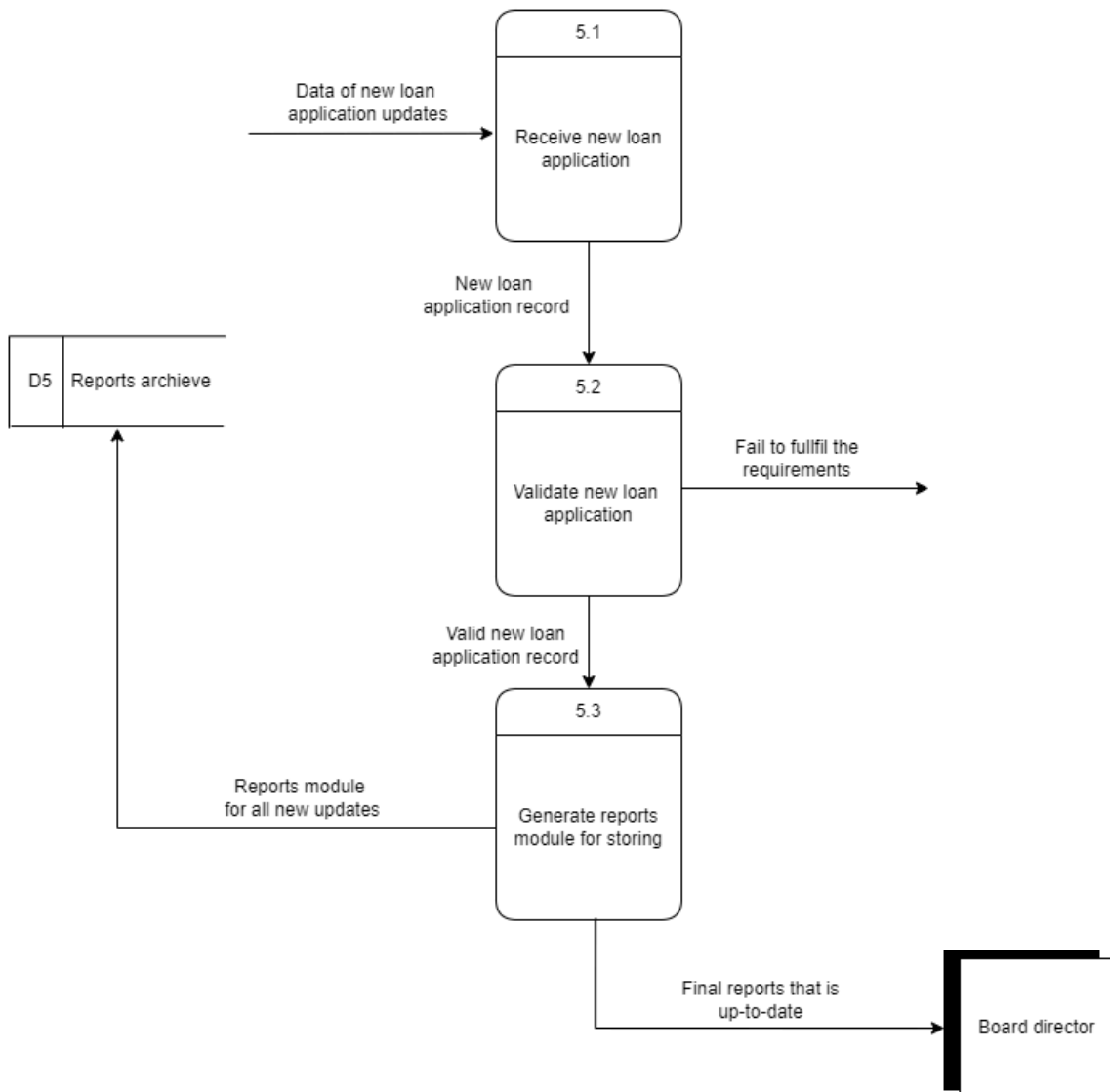
3.0 Saving and withdrawing money from account



4.0 User apply for loan (Loan Processing System)

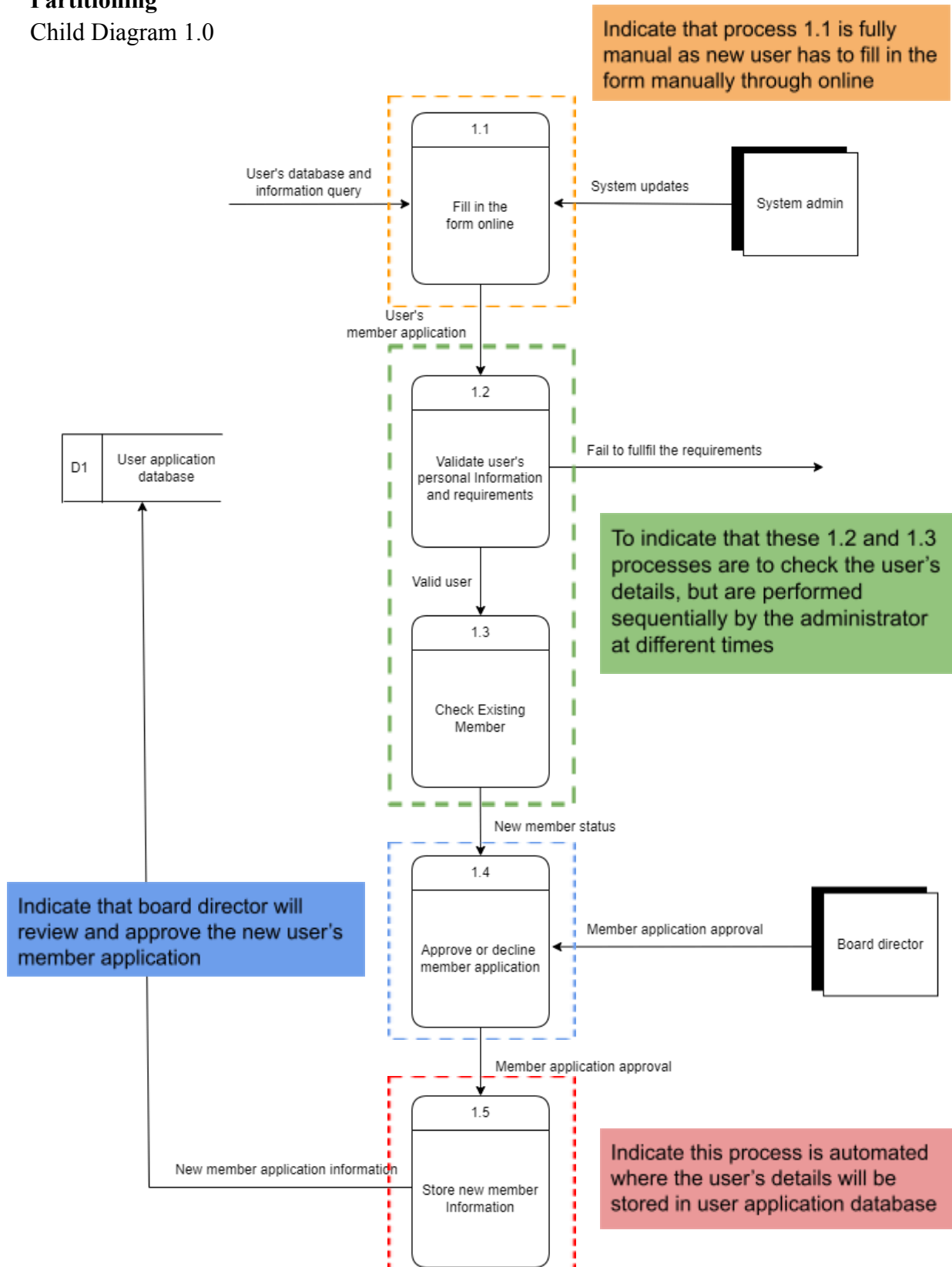


5.0 Reporting and finalize the new applications and financial modules

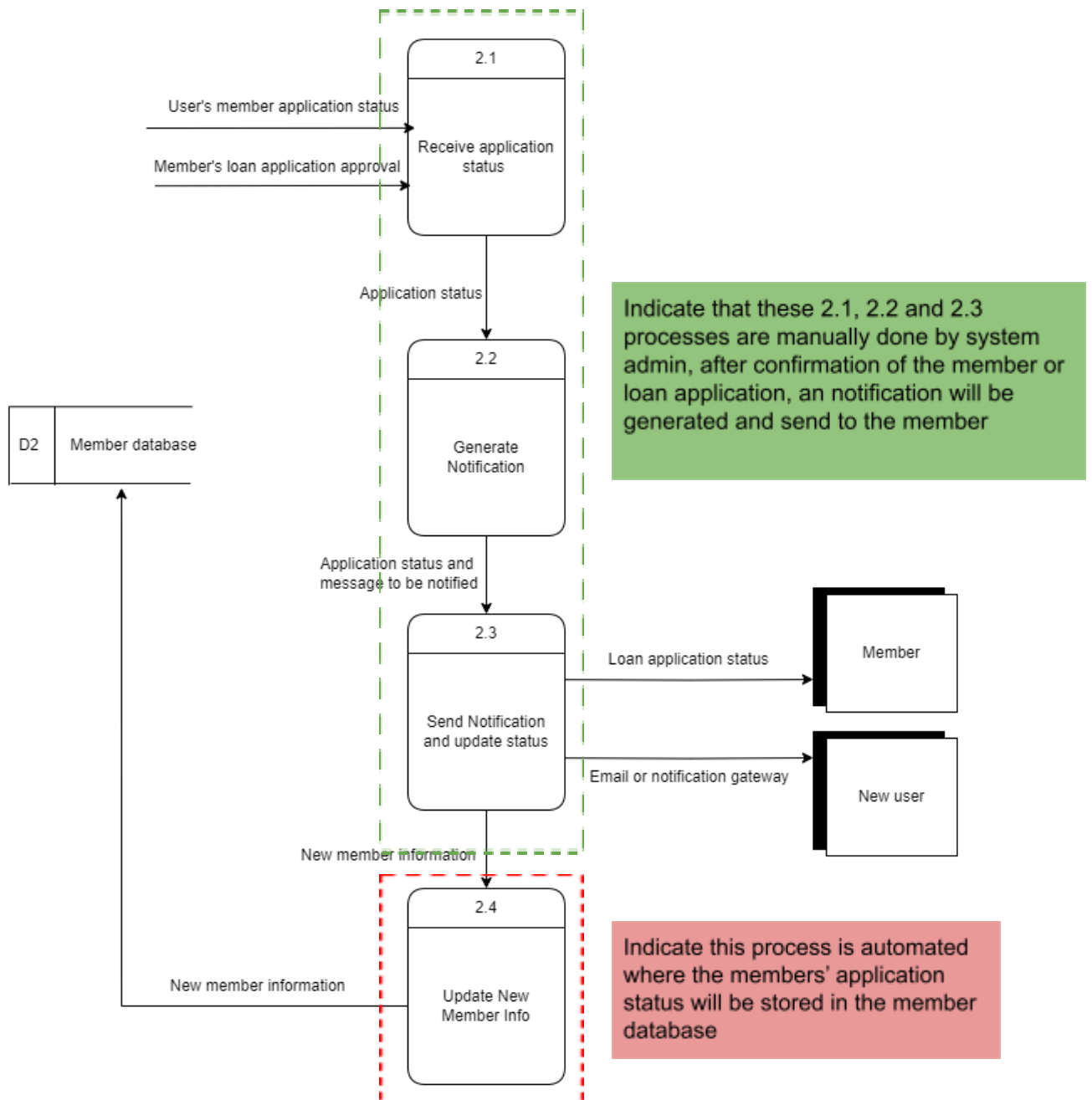


Partitioning

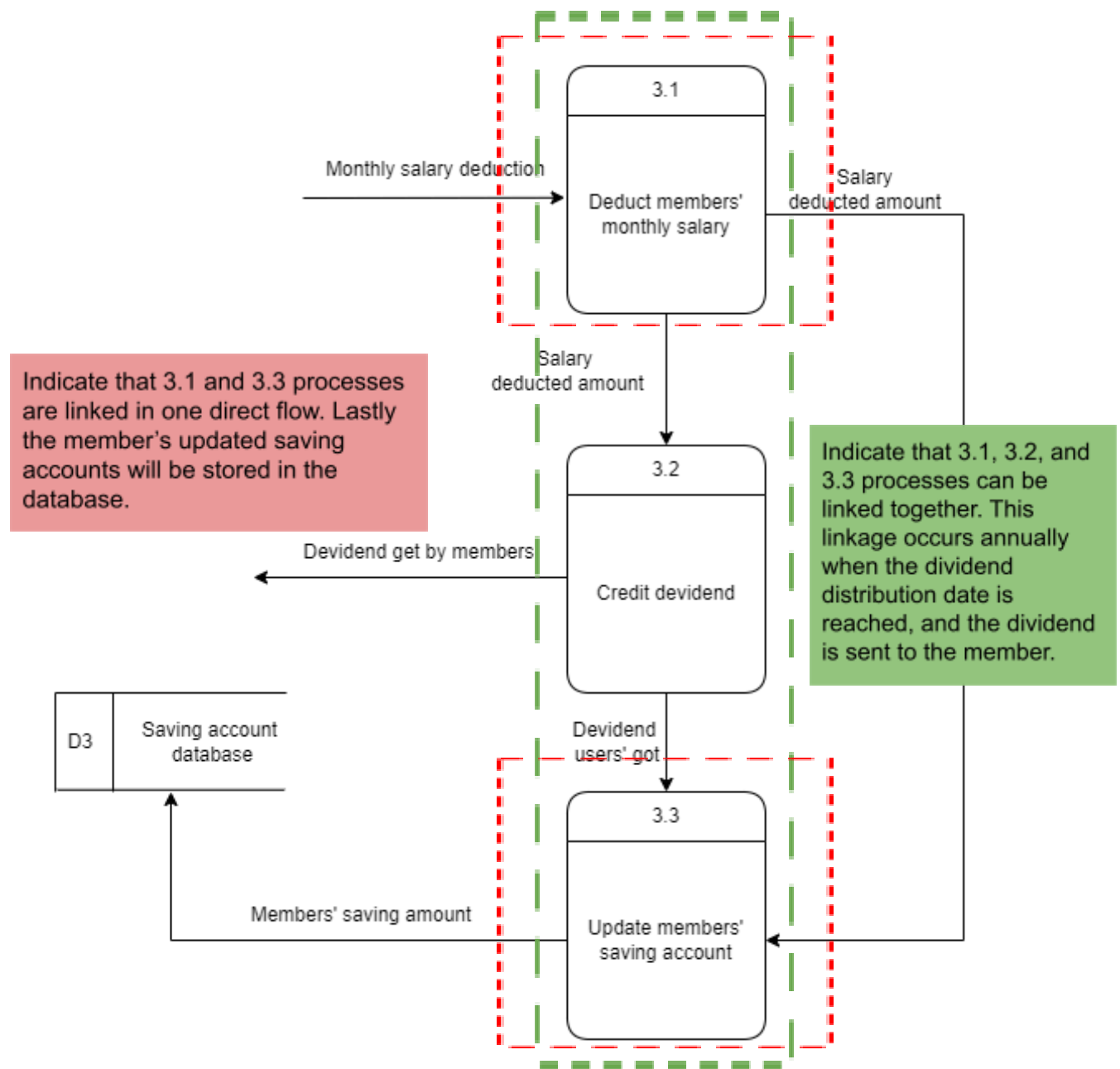
Child Diagram 1.0



Child Diagram 2.0

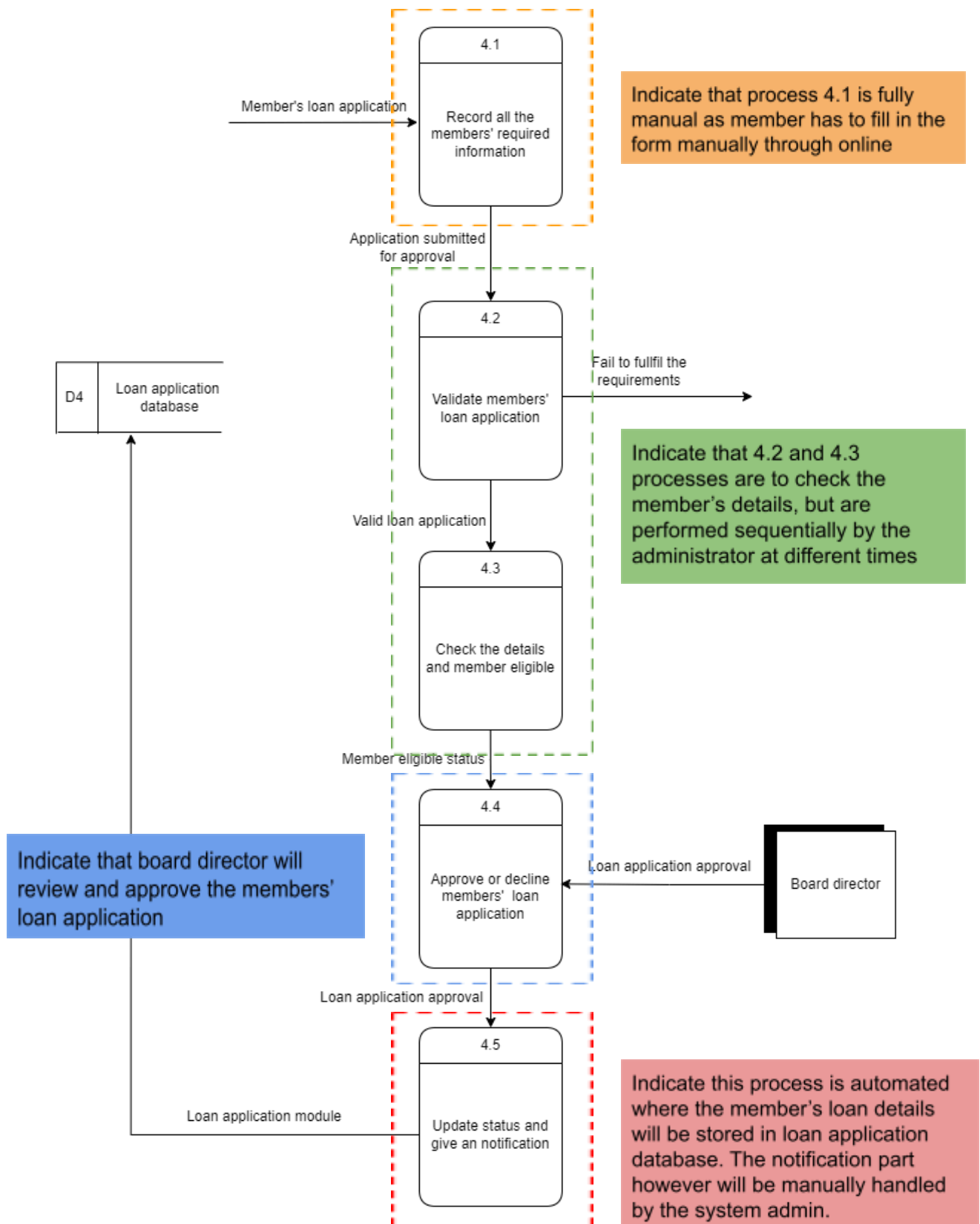


Child Diagram 3.0

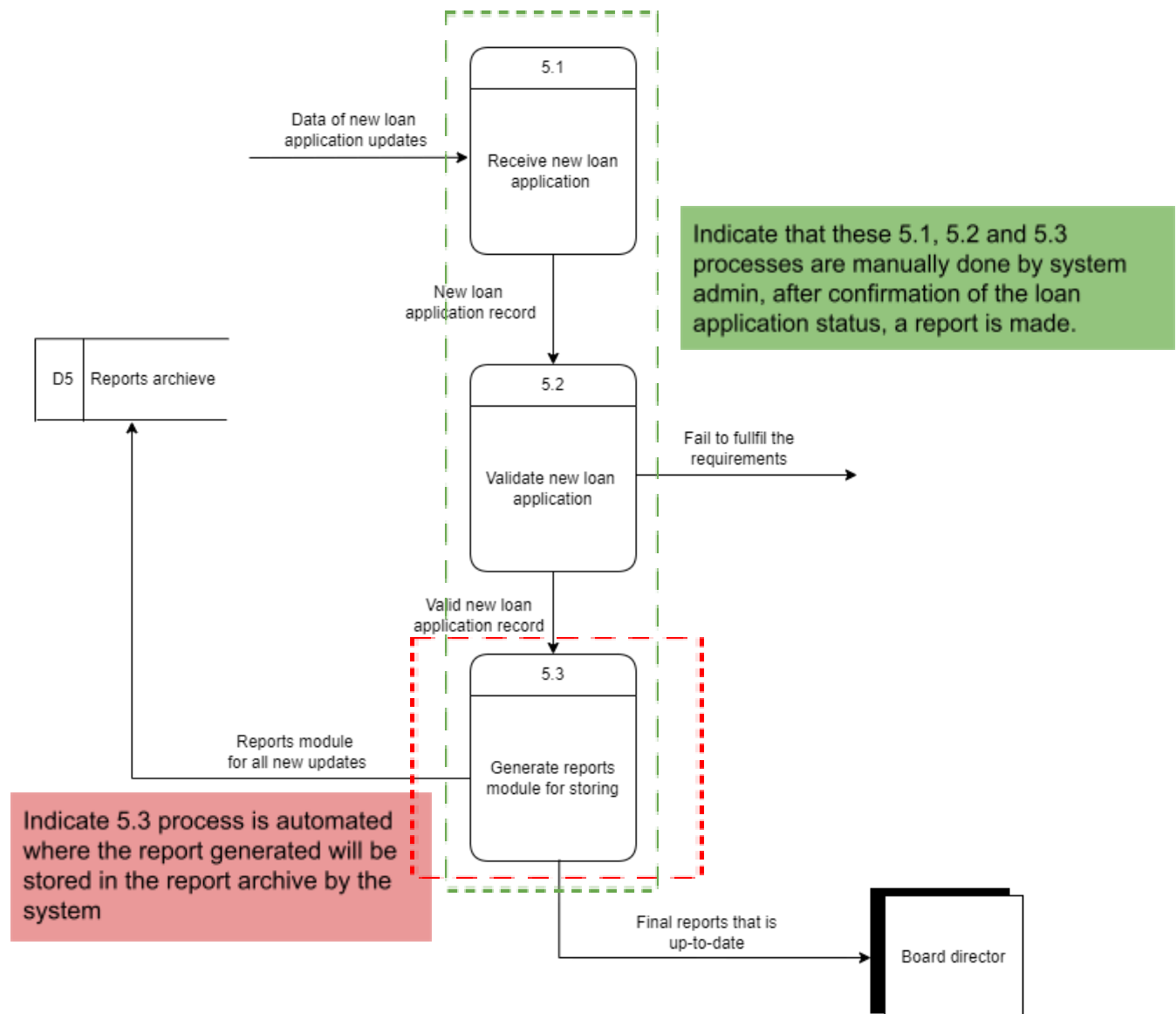


Although the processes are listed here, the actual workflow for specific payment calculations and corporate payment methods falls outside the system's scope. The banking aspects are managed by external banking companies.

Child Diagram 4.0



Child Diagram 5.0



CRUD Matrix

Activity	New User	Member	System Admin	Board Director	System	User Application Database	Member Database	Member Saving Account Database	Loan Application Database	Report Archive
Fill form	C					U				
Personal information validation			RU							
Check existing member			R				R			
Member application approval				RU			RU			
Store new member information list					U	UD	U			
Receive application status			R							
Generate notification			C							

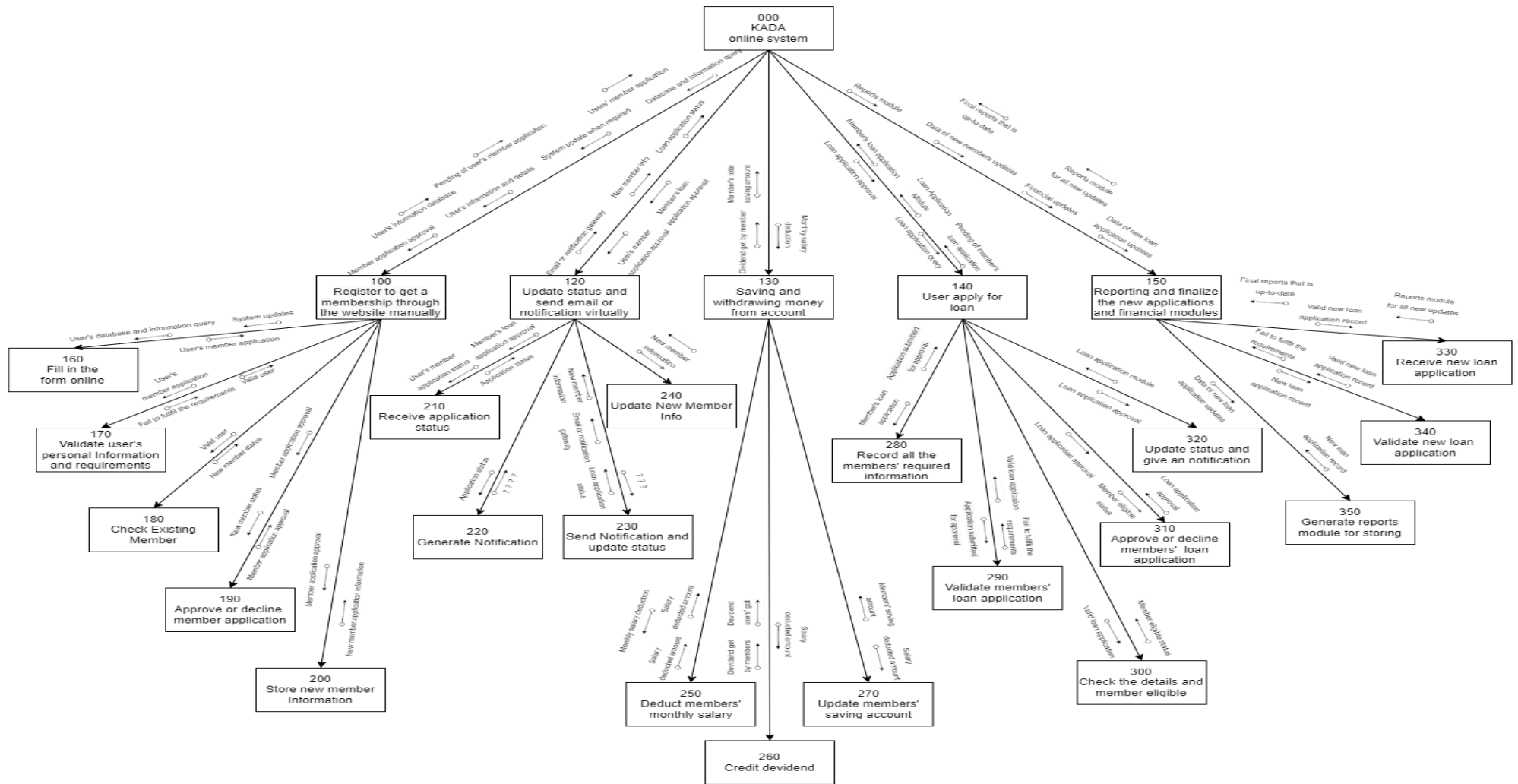
Send notification	R	R	U							
Update member information list					U		U			
Deduct monthly salary			RU					RU		
Credit dividend			CRU					RU		
Update saving account list					U			U		
Receive loan application		C	R						RU	
Validate loan application			RU							
Check member details and eligibility			R						R	
Loan application approval				RU						

Receive loan application		R								
Update the loan application list					U				U	
Generate report			C							RU
Update report archive list					U					U

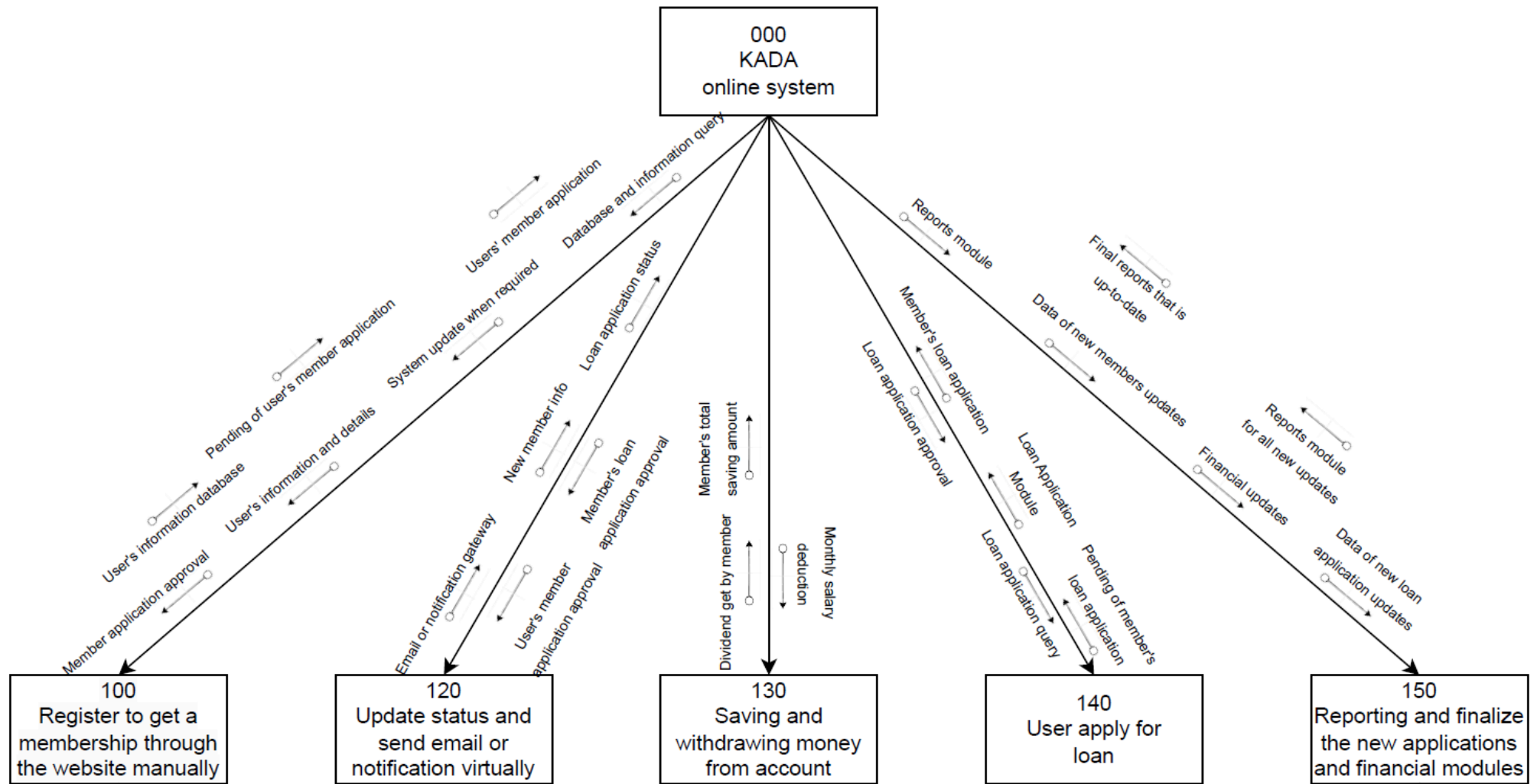
Event Response Table

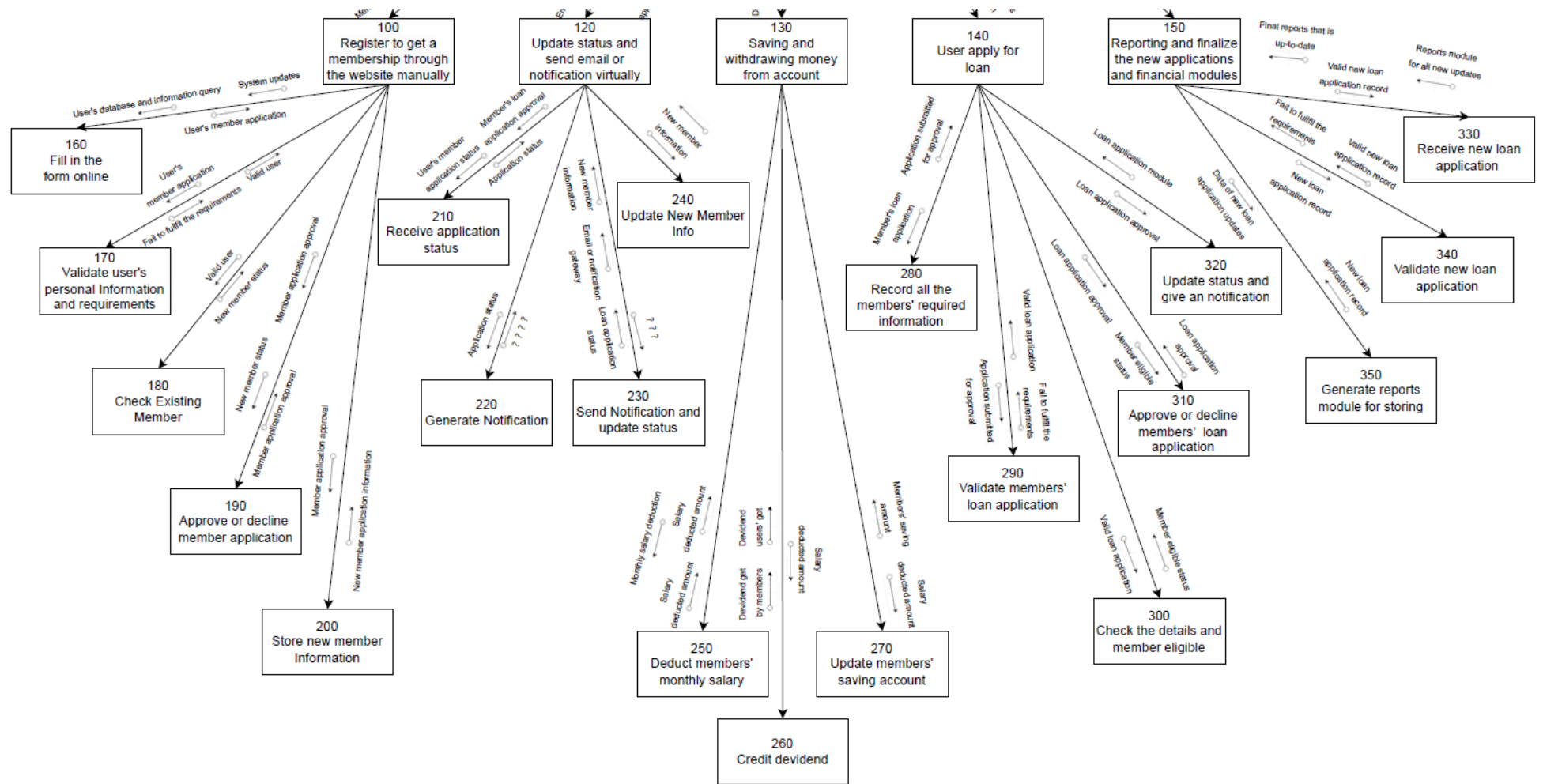
Event	Source	Trigger	Activity	Response	Destination
User submits their information	New User	Personal information	Validate personal information, store user information and generate member application data	User information	Member application data (D1)
Board director approved the member's application	Board director	Member Application	Update member application status, store new member information(D2) and admin notify the user about the approval status.	Member application status	Member data (D2)
Member saving	Member	Monthly salary deduction	Update saving account data (D3), store saving amount, update dividend info	Saving information	Saving account data(D3)
Member applies for a loan	Member	Loan application	Validate loan application, store loan account data(D4), and send the loan application to the board director for approval	Loan detail	Loan account data(D4)
Board director approved the loan application	Board director	Loan application approval	Update loan application status, store loan account data(D4), admin notify member about the approval status	Loan application status	Member
Board director views the annual report	Member application data (D1), Member data (D2), Saving account data(D3), Loan account data(D4)	Data output	Generate annual report, update the report data(D5), and send the report to the board director	Report update	Board director

Structure Chart



(in details)





System Architecture

